

JOONWON LEE

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SUMMARY

Computational statistics Ph.D. candidate with experience developing scalable statistical and machine learning workflows for large, complex datasets. Builds reproducible Python/PyTorch and LightGBM pipelines for predictive modeling, simulation, optimization, and model validation; combines rigorous statistical evaluation with clear recommendations for business stakeholders.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Computing: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM, CPU/GPU computing; SQL/MySQL (working familiarity)

Machine Learning and Statistics: Predictive modeling, GLM/GBM, feature attribution, causal inference, fixed-effects regression, time-series and spatio-temporal modeling, hypothesis testing, Monte Carlo simulation, numerical optimization, model selection and validation, uncertainty quantification

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring question into explicit hypotheses, likelihood-ratio boundaries, and statistically controlled decisions.
- Designed simulation and dynamic-programming evaluations of Type I/II error, detection delay, stopping-time uncertainty, and early-decision tradeoffs; documented model behavior and limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pipeline across 2.46M+ property-risk records, including data preparation, feature processing, GLM benchmarking, predictive-performance evaluation, and model validation on AWS EC2.
- Analyzed model performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

SELECTED RESEARCH

Ph.D. Dissertation Research - Scalable Statistical Computing and Model Diagnostics

Sep 2024 - Present

- **Scalable Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable space-time Gaussian processes, using ordered local conditioning for parameter estimation and uncertainty quantification with up to 145,008 observations per day.
- **Data and Model Diagnostics:** Built quality-filtering, missingness, time-dependent alignment, and irregular-grid matching workflows; developed scale- and frequency-resolved diagnostics to localize model misspecification and acquisition artifacts.
- **Research Engineering:** Implemented reusable Python/PyTorch workflows with L-BFGS optimization, CPU/GPU and restartable HPC execution, and performance-critical C++ routines exposed through pybind11.

Longitudinal Observational Study - EITC and Household Labor Supply

2021

- Estimated the effect of the Earned Income Tax Credit on household labor supply using fixed-effects regression and propensity-score matching with observed household demographic and economic covariates.

ADDITIONAL

Coursework: Machine Learning, Data Structures and Algorithms, Probability Theory, Statistical Computing, Stochastic Processes, Advanced Theory of Statistics I-II, Linear Algebra, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)